PROJECT PROPOSAL

Paddy Land Watch

AI that tracks what happens to every paddy tract each season, and warns early when paddy land is filled in, using Sentinel-1 radar and Sentinel-2 images

Wet zone: Gampaha, Colombo and Kalutara · Group 6, JARVIS · IS4990, October 2026

MEMBER 1
AI land-state model

MEMBER 2
Radar change detection

MEMBER 3
Pipeline and dashboard

The problem: paddy land is lost before anyone sees it

The law

Paddy land must be farmed every season (s. 30). Filling it without written permission is an offence: a fine up to Rs 100,000 or up to six months in prison (s. 32).

The scale

Over 100,000 acres of paddy lay uncultivated in a 2025 department survey. A 2018 audit in Kegalle found 372 filling cases and no legal action.

Today

Paper registers, field visits and complaints: officers find the filling after the soil is in.

Today

Soil is dumped → Weeks pass → A building starts → Someone complains

With Paddy Land Watch

Soil is dumped → Radar sees the change → Alert in 1–2 passes → An officer checks

Agrarian Development Act No. 46 of 2000; Auditor General, Kegalle District special audit, 2018.

Past research and the gap

Study	Where	What it did	What it leaves out
Özdoğan et al. 2025	All Sri Lanka, 2000–2022	Rice area and yield from Landsat and Sentinel-2; about 91–93% accurate in 2021	Photos only; "rice or not"; no idle or filled class
Sankavi et al. 2025, University of Moratuwa	North Central Province	A CNN finds paddy; NDVI and NDWI split active and inactive fields	Dry zone; no other-crop or filled class
Wanninayaka et al. 2020, KDU	Ampara	A Sentinel-1 time series maps paddy area	Paddy area only
Jayasooriya et al. 2023, KDU	Hambantota	Landsat 1990–2022 shows paddy lost near the Southern Expressway	Looks back; no alerts
Kasture et al. 2025, EGU	Udupi, India	Sentinel-1 radar with LSTM and BiGRU models finds abandoned paddy	Two classes only; not Sri Lanka



Our gap: no study sorts Sri Lankan paddy land into the four states the law and the subsidy use, every season, in the cloudy wet zone, with field-checked alerts.

Who uses it, and why it matters

The Department of Agrarian Development: 565 Agrarian Service Centres under 25 district offices (2023 report).

Agrarian Service Centre officers

They get

A monthly ranked list of likely filling, with before-and-after pictures

They

Visit the top cases, check the permit, act early

Subsidy officers

They get

Each tract's state this season

They

Check Rs 25,000/ha paddy and Rs 15,000/ha other-crop claims (Yala 2026)

Idle-land recovery

They get

Tracts idle for two or more seasons

They

Ask owners to farm them again

Census and Statistics

They get

A sown-area estimate for each district

They

Compare it with survey figures

An alert to check, never a verdict: a satellite sees the change; only an officer can say whether the filling was legal.

Four land states, four seasonal patterns

Paddy grown

From space: The field floods at planting (radar goes dark), the crop greens up, then it is harvested

Why it matters: Paddy subsidy, Rs 25,000/ha

Other crop on paddy land

From space: No standing water; a shorter green-up from maize, vegetables or similar

Why it matters: Other-crop subsidy, Rs 15,000/ha

Idle

From space: No flooding and no crop cycle; weeds or grass give a flat signal

Why it matters: Target for idle-land recovery

Filled or built over

From space: A sudden, lasting change to bare soil or roofs; the field never floods again

Why it matters: Possible offence under s. 32: inspect quickly

Radar sees the pattern through monsoon cloud. Reading several seasons at once tells a one-off fallow season from abandoned land.

Member 1: AI land-state model

RQ1 RQ3

How accurately can deep learning on fused Sentinel-1 and Sentinel-2 time series sort each tract's season into paddy, other crop, idle or filled?

Data analysis

- Label about 400 tract-seasons from field visits and Sentinel-2
- Score each class with F1 and macro-F1 in areas the model never saw
- Beat a Random Forest, AlphaEarth embeddings and the Özdoğan rice maps
- RQ3: accuracy when clear photos are few, and one-off fallow against abandoned land

Technical work

- Time-series deep learning: a temporal CNN or a transformer
- Inputs: Sentinel-1 VV and VH; Sentinel-2 B2, B3, B4, B8, B11, B12, NDVI and NDWI
- Masked attention that leans on radar on cloudy dates
- Reads several seasons at once

Output: a four-state map for every tract, every season

Member 2: Radar change detection and checks

RQ2 RQ3 NISAR stretch

How soon after filling starts can change detection flag it, and how many of the top 20 and top 50 alerts are real?

Data analysis

- Days from the start of filling to the alert
- Share of the top 20 and top 50 alerts confirmed in the field
- 30–50 known filling cases to test on
- Compare with a simple before-and-after change in radar and NDVI

Technical work

- Sentinel-1 calibration and terrain correction
- VV/VH ratio and change vector analysis, with no training data
- A radar stability index: filling against harvest or clearing
- Stretch: does NISAR L-band radar separate idle land from other crops?

Output: filling alerts within one or two radar passes, checked in the field

Member 3: Pipeline, ranking and dashboard

RQ4 Which way of ranking alerts gives the most confirmed cases per field visit?

Data analysis

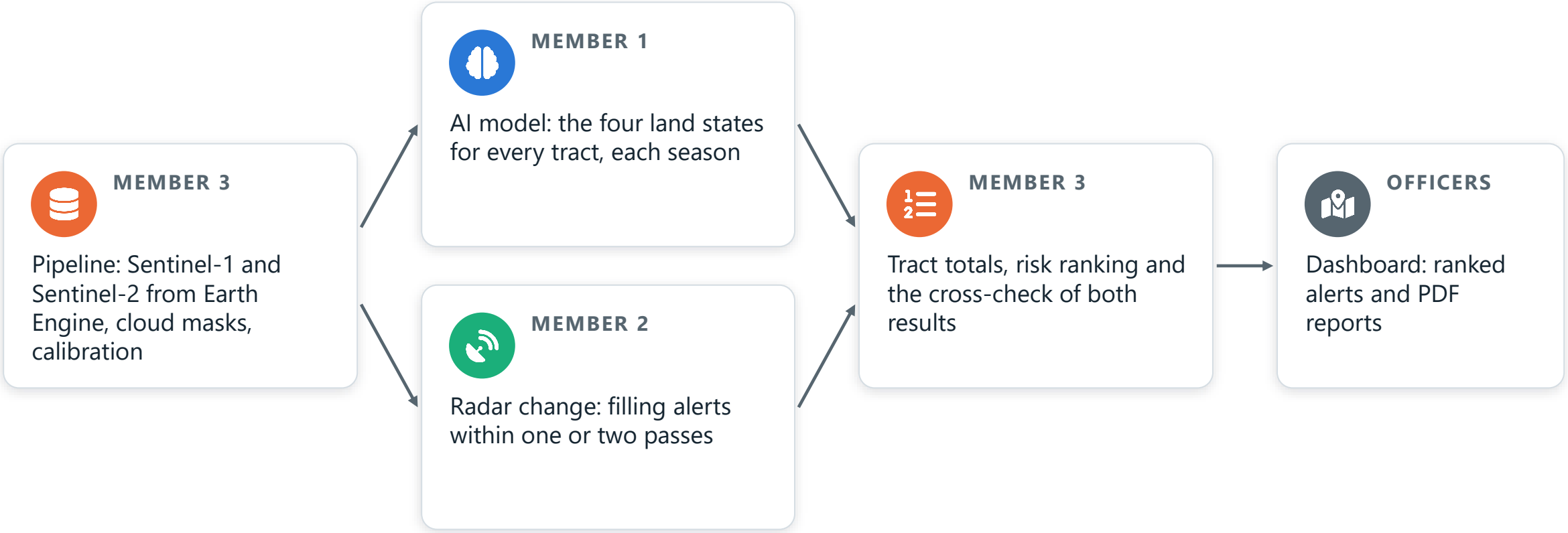
- Rank alerts by area, speed of change, nearness to roads and waterways, and past cases
- Confirmed cases per field visit, against ranking by area only
- District totals against Census and Statistics figures

Technical work

- Earth Engine to Python pipeline: download, cloud masks, co-registration
- Pixel results summed to each paddy tract (yaya)
- Multi-criteria risk ranking of tracts
- Web dashboard with alert lists and PDF reports for officers

Output: a ranked alert list and a dashboard for Agrarian Service Centre officers

One system, three parts



Feedback: every case an officer confirms becomes a new training label, so the model is retrained every season.

Data and ground truth

All images are free. Only the department's own lists must be asked for, so that request goes out in week one.

What	Source	Note
Radar time series	Sentinel-1, 10 m, sees through cloud (Earth Engine)	Gampaha got 9 scenes in Jul–Sep 2026, mostly 12 days apart
Optical time series	Sentinel-2, 10 m, about every 5 days	Many gaps under monsoon cloud
Paddy mask and baseline	Özdoğan et al. seasonal rice maps and code	Which land has grown rice, season by season
Few-label baseline	AlphaEarth Satellite Embedding, 10 m, 2017–2024	Does a simple model on embeddings match deep learning?
District totals	Census and Statistics, Maha and Yala	A check on totals, not on tracts
Field labels	About 400 tract-seasons in Gampaha and Kalutara, Maha 2026/27	GPS phone photos from team visits
Department records	Uncultivated-land list, fallow registers, filling permits	Not public: request with a supervisor's letter

Feasibility, targets and what we deliver

Why it is feasible

- Free data: Sentinel-1, Sentinel-2 and Earth Engine
- Python and PyTorch
- Maha 2026/27 is the live season for field checks
- Week-6 check before the main model work

Targets

- Week 6: macro-F1 of 0.70 or more
- End: four-state accuracy above 85%
- Filling alert within 12–24 days (one or two radar passes)
- Fewer than 15% false alerts in the top list

What we deliver

- Four-state maps every season
- A radar filling-alert engine
- An automatic pipeline
- A web dashboard for officers
- A research paper

Biggest risks: idle and other crop may look alike (then merge them as "not paddy"); the department may not share its lists (then use field labels only).

Thank you

Questions and feedback are welcome.

Datasets and links

- Özdoğan et al.: Sri Lanka rice maps and code
- Sentinel-1 radar in Earth Engine
- Sentinel-2 in Earth Engine
- AlphaEarth Satellite Embedding in Earth Engine
- Copernicus Browser: see any Sentinel picture
- Census and Statistics: paddy statistics
- Kasture et al. 2025: abandoned paddy from radar (EGU abstract)

Sources

- Agrarian Development Act No. 46 of 2000, sections 30 and 32
- Department of Agrarian Development, Annual Performance Report 2023
- Auditor General, Kegalle District special audit, 2018
- News 1st, 27 Feb 2025: uncultivated paddy land
- Ada Derana, 4 Mar 2026: Yala 2026 subsidies